

Code Repository Overview — DentaCoPilot

Open-source pipeline accompanying the BigMouth project proposal

Carson Conception Rodrigues (CELABE)

Steffie Dione Rebello (KLE Vishwanath Katti Institute of Dental Sciences)

Repository [TODO: insert public Git URL when published] (*MIT licence; private during development, public at submission.*)

Quick orientation

The DentaCoPilot codebase implements every model evaluated in the manuscript, the synthetic chart generator that produced the corpus, and the audit / replay infrastructure that makes the run records reviewable.

Layout

```
07-dental-procedure-prediction/
|-- draft/paper.tex          full manuscript
|-- references/references.bib 46-entry bibliography
|-- figures/                  fig1..fig5 (PDF + PNG)
|-- data/
|   |-- cc_compute.py         Claude Code subprocess wrapper (single chokepoint)
|   |-- cc_compute_test.py    smoke test
|   |-- cdt_codes.py          CDT 2024 vocabulary + category map
|   |-- tooth.py              Universal/FDI tooth-numbering helpers
|   |-- synth_charts.py        synthetic chart generator (5 archetypes)
|   |-- evaluate.py            single CLI entry point for all experiments
|   |-- calibration.py         temperature scaling, ECE, reliability curves
|   |-- abstention.py          coverage-risk, threshold selection
|   |-- hybrid_analysis.py     confidence-routed gating sweep
|   |-- retrieval.py           BM25 over the open evidence corpus
|   |-- evidence_corpus/       20 paraphrased clinical-practice cards
|   |-- loaders/               DENTEX, MMDental, DenPAR public-data loaders
|   |-- models/
|       |-- base.py            Model interface
|       |-- b0_bigram.py        B0 frequency bigram
|       |-- b1_tfidf_lr.py      B1 TF-IDF + LR
|       |-- b2_xgboost.py       B2 XGBoost
|       |-- b3_multitp.py       B3 MultiTP-style CNN-RNN
|       |-- llm_base.py         shared LLM scaffolding
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| | | -- m1_haiku.py           M1
| | | -- m2_sonnet_cot.py      M2
| | | -- m3_sonnet_rag.py      M3
| | | -- m4_opus.py           M4
| | | -- m5_sonnet_baseline_primed.py  M5 (proposed forward design)
| | | -- m6_opus_baseline_primed.py    M6
| | -- requirements.txt        pinned, anthropic SDK deliberately absent
| | -- README.md
| | -- results_*/              every experimental run record
|-- notes/                     planning, IRB, BigMouth proposal, preregistration
|-- submission/               medRxiv / arXiv / OSF submission bundles
|-- Makefile                  smoke / full / pdf / docx / verify / finalize
|-- build.sh                  self-contained PDF + DOCX builder (per-paper)

```

How a reviewer can audit our work in 30 minutes

1. Clone the repository.
2. `cd 07-dental-procedure-prediction && make smoke` runs the full Stage-1 pipeline (CDT vocabulary → synthetic chart generation → B0/B1/B2 baselines → figures) end-to-end without spending any Claude Code quota.
3. Inspect any `data/results_<run>/llm_calls.jsonl` — every LLM call is logged with full prompt, full response, latency, and token counts. The reviewer can read the model’s reasoning on any individual case without re-running anything.
4. `make verify` runs the no-anthropic-SDK CI check (the grep guard ensuring no metered API path can be introduced).

Reproducibility primitives

- **Seeds:** `random.seed(42)`, `numpy.random.seed(42)`, `torch.manual_seed(42)` at the top of every script.
- **Run records:** `config.json` (git SHA + seed + CLI args + Python version + platform), `results.json` (per-trial predictions), `summary.json` (aggregated metrics), `llm_calls.jsonl` (every LLM call).
- **Determinism:** re-running a baseline with the same seed produces byte-identical `summary.json`; LLM models reproduce within stochastic tolerance documented in the manuscript.

Compute substrate

A central design choice: every LLM inference call goes through the local Anthropic Claude Code CLI rather than the metered Anthropic API. The single subprocess wrapper at `data/cc_compute.py` invokes `claude -p ... --output-format json` and parses the structured response. The full call record is logged to a per-run JSONL audit trail.

What we will share with BigMouth

- Read access to the live private repository for the BigMouth project review committee, on request.
 - A short live walk-through (videoconference) of any module the committee wants to inspect, on request.
 - Public open-source release on the project's accepted-paper publication date; MIT licence; no patient data in the repository.
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For repository questions, contact carson@celabe.com.